

T0-Theory: Cosmic Relations

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Abstract

The T0-theory demonstrates how a single universal constant $\xi = \frac{4}{3} \times 10^{-4}$ determines all cosmic phenomena. This document presents the fundamental relationships between the gravitational constant,

cosmic microwave background radiation (CMB), Casimir effect and cosmic structures within the framework of a static, eternally existing universe. All derivations are performed in natural units ($\hbar = c = k_B = 1$) and respect the time-energy duality as a fundamental principle of quantum mechanics.

.1 Introduction: The Universal ξ -Constant

Foundations of T0 Theory

Important

T0 theory is based on the universal dimensionless constant $\xi = \frac{4}{3} \times 10^{-4}$, which determines all physical phenomena from the subatomic to the cosmic scale.

T0 theory revolutionizes our understanding of the universe through the introduction of a single fundamental constant. This constant forms the basis for all physical calculations and predictions of the theory:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333333... \times 10^{-4} \quad (1)$$

This dimensionless constant connects quantum and gravitational phenomena, enabling a unified description of all fundamental interactions.

Note on Derivation

For the detailed derivation and physical justification of this fundamental constant, see the document "Parameter Derivation" (available at:).

Time-Energy Duality as Foundation

Revolutionary

Heisenberg's uncertainty relation $\Delta E \times \Delta t \geq \hbar/2 = 1/2$ (natural units) provides irrefutable proof that a classical Big Bang singularity with infinite density is physically impossible and is replaced by a tiny but finite core with minimal length scale L_0 (from ξ).

Heisenberg's uncertainty relation between energy and time represents the fundamental principle of T0-theory:

$$\Delta E \times \Delta t \geq \frac{1}{2} \quad (\text{natural units}) \quad (2)$$

This relation has far-reaching cosmological consequences:

- A temporal beginning (Big Bang) would mean $\Delta t = \text{finite}$
- This leads to $\Delta E \rightarrow \infty$ - physically inconsistent
- Therefore the universe must have existed eternally: $\Delta t = \infty$
- The universe is static, without expanding space

.2 Cosmic Microwave Background (CMB)

CMB without Big Bang: ξ -Field Mechanisms

Revolutionary

Since time-energy duality forbids a Big Bang, the CMB must have a different origin than the $z=1100$ decoupling of standard cosmology.

T0-theory explains the CMB through ξ -field quantum fluctuations:

$$\frac{T_{\text{CMB}}}{E_{\xi}} = \frac{16}{9} \xi^2 \quad (3)$$

With $E_{\xi} = \frac{1}{\xi} = \frac{3}{4} \times 10^4$ (natural units) and $\xi = \frac{4}{3} \times 10^{-4}$ this yields:

$$T_{\text{CMB}} = \frac{16}{9} \xi^2 \times E_{\xi} = \frac{16}{9} \times 1.78 \times 10^{-8} \times 7500 = 2.35 \times 10^{-4} \quad (4)$$

Conversion to SI units:

$$T_{\text{CMB}} = 2.725 \text{ K} \quad (5)$$

This agrees perfectly with observations!

CMB Energy Density and ξ -Length Scale

The CMB energy density in natural units is:

$$\rho_{\text{CMB}} = 4.87 \times 10^{41} \quad (\text{natural units, dimension } [E^4]) \quad (6)$$

This energy density defines a characteristic ξ -length scale:

$$L_\xi = \left(\frac{\xi}{\rho_{\text{CMB}}} \right)^{1/4} \quad (7)$$

Fundamental relation of CMB energy density:

$$\rho_{\text{CMB}} = \frac{\xi}{L_\xi^4} = \frac{\frac{4}{3} \times 10^{-4}}{(L_\xi)^4} \quad (8)$$

.3 Casimir Effect and ξ -Field Connection**Casimir-CMB Ratio as Experimental Confirmation****Experiment**

The ratio between Casimir energy density and CMB energy density confirms the characteristic ξ -length scale of $L_\xi = 10^{-4}$ m.

The Casimir energy density at plate separation $d = L_\xi$ is:

$$|\rho_{\text{Casimir}}| = \frac{\pi^2}{240 \times L_\xi^4} \quad (\text{natural units}) \quad (9)$$

The experimental ratio yields:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = \frac{\pi^2}{240\xi} = \frac{\pi^2 \times 10^4}{320} \approx 308 \quad (10)$$

Experimental confirmation: With $L_\xi = 10^{-4}$ m, direct calculation gives:

$$|\rho_{\text{Casimir}}| = \frac{\hbar c \pi^2}{240 \times (10^{-4})^4} = 1.3 \times 10^{-11} \text{ J/m}^3 \quad (11)$$

$$\rho_{\text{CMB}} = 4.17 \times 10^{-14} \text{ J/m}^3 \quad (12)$$

$$\text{Ratio} = \frac{1.3 \times 10^{-11}}{4.17 \times 10^{-14}} = 312 \quad (13)$$

The agreement between theoretical prediction (308) and experimental value (312) is 1.3% - excellent confirmation!

ξ -Field as Universal Vacuum

Important

The ξ -field manifests both in free CMB radiation and in geometrically constrained Casimir vacuum. This proves the fundamental reality of the ξ -field.

The characteristic ξ -length scale L_ξ is the point where CMB vacuum energy density and Casimir energy density reach comparable magnitudes:

$$\text{Free vacuum: } \rho_{\text{CMB}} = +4.87 \times 10^{41} \quad (14)$$

$$\text{Constrained vacuum: } |\rho_{\text{Casimir}}| = \frac{\pi^2}{240d^4} \quad (15)$$

.4 Cosmic Redshift without Expansion

ξ -Field Energy Loss Mechanism

Revolutionary

The observed cosmic redshift arises not from spatial expansion but from energy loss of photons in the omnipresent ξ -field.

Photons lose energy through interaction with the ξ -field:

$$\frac{dE}{dx} = -\xi \cdot f\left(\frac{E}{E_\xi}\right) \cdot E \quad (16)$$

For the linear case $f\left(\frac{E}{E_\xi}\right) = \frac{E}{E_\xi}$ this yields:

$$\frac{dE}{dx} = -\frac{\xi E^2}{E_\xi} \quad (17)$$

Wavelength-Dependent Redshift

Integration of the energy loss equation leads to wavelength-dependent redshift:

Wavelength-dependent redshift:

$$z(\lambda_0) = \frac{\xi x}{E_\xi} \cdot \lambda_0 \quad (18)$$

where λ_0 is the emitted wavelength and x is the distance traveled.

This formula predicts:

- Shorter wavelength light (UV) shows greater redshift
- Longer wavelength light (radio) shows smaller redshift
- The ratio is $z_1/z_2 = \lambda_1/\lambda_2$

Experiment

Experimental test: Comparison of radio and optical redshifts

- 21cm hydrogen line: $\nu = 1420$ MHz
- Optical $H\alpha$ line: $\nu = 457$ THz
- Predicted ratio: $z_{21\text{cm}}/z_{H\alpha} = 3.1 \times 10^{-6}$

.5 Structure Formation in the Static ξ -Universe

Continuous Structure Development

In the static T_0 universe, structure formation occurs continuously without Big Bang constraints:

$$\frac{d\rho}{dt} = -\nabla \cdot (\rho \mathbf{v}) + S_{\xi}(\rho, T, \xi) \quad (19)$$

where S_{ξ} is the ξ -field source term for continuous matter/energy transformation.

ξ -Supported Continuous Creation

The ξ -field enables continuous matter/energy transformation:

$$\text{Quantum vacuum} \xrightarrow{\xi} \text{Virtual particles} \quad (20)$$

$$\text{Virtual particles} \xrightarrow{\xi^2} \text{Real particles} \quad (21)$$

$$\text{Real particles} \xrightarrow{\xi^3} \text{Atomic nuclei} \quad (22)$$

$$\text{Atomic nuclei} \xrightarrow{\text{Time}} \text{Stars, galaxies} \quad (23)$$

Energy balance is maintained by:

$$\rho_{\text{total}} = \rho_{\text{matter}} + \rho_{\xi\text{-field}} = \text{constant} \quad (24)$$

.6 Dimensionless ξ -Hierarchy

Energy Scale Ratios

All ξ -relations reduce to exact mathematical ratios:

Table 1: Dimensionless ξ -ratios

Ratio	Expression	Value
Temperature	$\frac{T_{\text{CMB}}}{E_{\xi}}$	3.13×10^{-8}
Theory	$\frac{16}{9} \xi^2$	3.16×10^{-8}
Length	$\frac{\ell_{\xi}}{L_{\xi}}$	$\xi^{-1/4}$
Casimir-CMB	$\frac{ \rho_{\text{Casimir}} }{\rho_{\text{CMB}}}$	$\frac{\pi^2 \times 10^4}{320}$

Important

All ξ -relations consist of exact mathematical ratios:

- Fractions: $\frac{4}{3}, \frac{3}{4}, \frac{16}{9}$
- Powers of ten: $10^{-4}, 10^3, 10^4$
- Mathematical constants: π^2

NO arbitrary decimal numbers! Everything follows from ξ -geometry.

.7 Experimental Predictions and Tests

Precision Measurements of Gravitational Constant

T0-theory predicts:

$$G_{T0} = 6.67430000... \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2) \quad (25)$$

This theoretically exact prediction can be tested by future precision measurements.

Casimir Force Anomalies

Experiment

Prediction: Casimir force anomalies at characteristic ξ -length scale

- Standard Casimir law: $F \propto d^{-4}$
- ξ -field modifications at $d = L_\xi = 10^{-4} \text{ m}$
- Measurable deviations through ξ -vacuum coupling

Electromagnetic Resonance

Maximum ξ -field-photon coupling at characteristic frequency:

$$\nu_\xi = \frac{1}{L_\xi} = 10^4 \text{ Hz} = 10 \text{ kHz} \quad (26)$$

Electromagnetic anomalies should occur at this frequency.

.8 Cosmological Consequences

Solution to Cosmological Problems

The T0 model solves all fine-tuning problems of standard cosmology:

Table 2: Cosmological problems: Standard vs. T0

Problem	Λ CDM	T0 Solution
Horizon problem	Inflation required	Infinite causal connect
Flatness problem	Fine-tuning	Geometry stabilizes ov time
Monopole problem	Topological defects	Defects dissipate ove time
Lithium problem	Nucleosynthesis dis- crepancy	Nucleosynthesis over time
Age problem	Objects older than uni- verse	Objects can be arbitrar
H_0 tension	9% discrepancy	No H_0 in static universe
Dark energy	69% of energy density	Not required

Parameter Reduction

Revolutionary

Revolutionary parameter reduction: From 25+ parameters to one!

- Standard model of particle physics: 19+ parameters
- Λ CDM cosmology: 6 parameters
- T0-theory: 1 parameter (ξ)

96% reduction!

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